

LUBBOCK, TX, USA

WMO: 722670

Lat: 33.666N

Lon: 101.823W

Elev: 992

StdP: 89.96

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23042

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-6.6	-18.1	0.9	-0.7	-15.8	1.1	2.9	15.0	11.5	13.6	13.1	4.4	30	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	37.6	19.0	36.3	19.4	35.0	19.4	22.5	31.0	21.9	30.5	21.4	29.9	6.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	16.9	24.9	19.7	16.3	24.4	19.1	15.7	24.2	71.3	30.7	69.0	30.5	67.0	29.9	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.1	11.7	10.5	DB	-12.6	40.4	2.4	2.2	-14.3	42.0	-15.7	43.3	-17.0	44.5	-18.7	46.1	(4)
			WB	-13.5	23.7	2.3	0.8	-15.1	24.3	-16.4	24.7	-17.7	25.1	-19.4	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	5.2	7.3	11.7	16.2	21.3	26.0	27.3	26.6	22.5	16.6	10.3	5.4
	DBStd	9.16	5.07	5.65	5.49	4.93	4.64	3.25	2.52	2.59	3.59	4.72	5.01	5.06
	HDD10.0	544	161	106	46	9	1	0	0	0	0	8	56	158
	HDD18.3	1787	406	308	213	94	24	1	0	0	9	88	242	402
	CDD10.0	2887	13	31	98	196	352	479	536	514	376	212	66	14
	CDD18.3	1088	0	0	7	31	117	230	278	256	134	34	2	0
	CDH23.3	12991	7	40	175	523	1459	2705	3329	2897	1373	421	57	5
CDH26.7	6469	0	6	41	183	719	1464	1795	1514	602	136	9	0	
Wind	WSAvg	5.3	5.2	5.5	6.0	6.4	6.0	5.9	4.8	4.4	4.4	4.9	5.0	5.0
Precipitation	PrecAvg	474	14	17	24	29	63	69	55	54	64	45	20	17
	PrecMax	846	70	64	151	147	308	215	183	225	221	274	169	57
	PrecMin	151	0	0	0	0	1	0	0	0	0	0	0	0
	PrecStd	140	16	16	27	29	51	46	45	44	57	53	28	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.3	27.4	30.2	32.9	38.0	40.5	39.1	38.3	36.1	32.6	27.9	23.8
		MCWB	9.9	11.7	13.0	13.8	16.5	18.1	19.6	20.0	19.3	16.4	13.1	10.2
	2%	DB	21.2	24.2	27.6	30.9	35.6	37.7	37.1	36.3	33.9	30.1	24.9	20.6
		MCWB	9.0	10.3	12.0	13.4	16.2	18.6	20.0	20.2	18.9	16.6	12.4	8.9
	5%	DB	18.0	21.4	25.2	28.9	33.3	35.8	35.8	35.1	32.3	28.0	22.5	17.6
		MCWB	7.6	9.5	11.4	13.3	16.1	19.1	20.1	20.2	19.0	16.0	11.6	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	15.1	18.0	22.4	26.5	30.9	34.0	34.4	33.7	30.3	25.6	19.7	14.8
		MCWB	6.4	8.0	10.8	13.0	16.1	19.2	20.2	20.3	18.9	15.2	10.7	7.0
	2%	WB	11.9	13.8	16.5	18.8	21.2	23.0	23.4	23.2	22.0	20.8	16.6	13.2
		MCDB	19.0	20.0	21.2	25.1	28.2	32.3	31.0	31.5	29.6	26.1	21.0	17.4
	5%	WB	10.0	11.9	15.1	17.3	20.1	22.0	22.4	22.5	21.3	19.4	14.7	10.6
		MCDB	18.5	20.1	20.9	23.0	27.5	30.7	30.9	31.0	28.6	24.8	20.1	17.7
Mean Daily Temperature Range	5% DB	WB	8.5	10.4	13.6	16.0	19.3	21.5	21.8	21.8	20.6	18.2	13.3	9.0
		MCDB	16.9	19.5	20.7	22.5	26.8	29.9	30.7	30.5	27.9	23.3	19.1	16.6
	5% WB	WB	6.9	8.9	11.9	14.8	18.5	20.8	21.3	21.3	20.0	17.1	11.9	7.3
		MCDB	14.6	17.7	20.5	22.0	25.8	29.1	30.1	29.9	27.2	22.3	18.3	14.5
	Clear-Sky Solar Irradiance	taub	0.260	0.271	0.297	0.331	0.342	0.382	0.387	0.374	0.347	0.317	0.288	0.267
		taud	2.580	2.560	2.496	2.390	2.424	2.355	2.391	2.386	2.443	2.503	2.517	2.556
Ebn at Noon		974	993	984	955	941	899	893	901	916	924	927	943	
Edh at Noon		77	88	102	119	116	125	120	118	106	92	82	75	
All-Sky Solar Radiation	RadAvg	3.34	4.12	5.38	6.60	7.08	7.53	7.22	6.64	5.50	4.37	3.48	2.89	
	RadStd	0.32	0.39	0.39	0.45	0.56	0.43	0.35	0.29	0.38	0.49	0.36	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	+73

Nomenclature: See separate page